INRUSH :: PROTO-TYPING A QUEER METHODOLOGY FOR NETWORKED PERFORMANCE

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ABSTRACT

Inrush is a collection of java classes and max patches, proto-typed by Dan Thorpe between 2014-2017 to explore the potential of queer-theory methodologies in reactive performances. The result, and what this paper reports on, is a toolbox for composing mixed graphic and traditional notation works, synchronised over network, and drawn on the fly using OSC messages timed through input either from analysis of a dataset, a "score performer", or a mixture of the two. The first focus of this paper is on the theoretical motivations of such an approach, and two case-studies of how Dan Thorpe has used this approach in two reactive works: firstly, a more in-depth study of his work *A20* (for piano, projection, generated traditional notation scrolling score and electronics), and a report on his work in progress *scène* (a double concertino for flute & electronics with mixed ensemble, electronics, and rehearsal-realised mixed notation score), which will be presented at this conference.

1. INTRODUCTION

Inrush is a collection of java classes and max patches I prototyped from 2014-2017. I began writing software with the aim of exploring reactive approaches to notation and performances that acknowledged and embraced performer body autonomy, explored ways of sensing bodies and musical gesture that celebrated ambivalence and error, and struck a middle ground between fixed and unfixed/composed and improvised. The result, and what this paper reports on, is a toolbox for composing mixed graphic and traditional notation works, synchronised over network, and drawn on the fly using OSC messages timed through input either from analysis of a dataset, a "score performer", or a mixture of the two.

This paper will focus on the theoretical motivations of such an approach, and two case-studies of how I have used this approach in two reactive works: firstly, a more in-depth study of my work *A20* (for piano, projection, generated traditional notation scrolling score and electronics), and a report on my work in progress *scène* (a double concertino for flute & electronics with mixed ensemble, electronics, and rehearsal-realised mixed notation score), which will be presented at this conference.

2. THEORETICAL MOTIVATIONS

2.1. Queering I/O

In his 2002 paper *Interactivity, where to from here?*, Paine raises an important question of semantics: why do we talk about systems that offer no possibilities beyond those pre-determined by their designer as *interactive?* These systems, as Paine terms it, are *reactive*: responding to performer choices but with no level of cognition. Indeed, most systems we currently term as interactive are re-categorised by this definition as reactive (Bongers, 2000).

On the face of it, this difference is purely semantic, but a technical language that indicates falsely that users have the power to shape their relationship with technology reflects more broadly on culture that insists on participation in ubiquitous institutions without any real power in their shaping or articulation. There is a queer necessity to imagine and enact possible futures outside of these limits, to explore the not-here-yet of which queerness offers us a glimpse. This project is, therefore, the beginning of a personal, queer resistance to this technological and social paradigm: an experiment in a reactivity that is by design editable, extensible and manipulable by users with even basic technological literacies. An experiment in the redistribution of the power of decision making, structuring and code-level control from the system designer and explore new relationships between designer and users and the documentation of methodologies to do so.

This desire for negotiability, dispersion of power and extensibility in reactive design and in performance comes from a radical queer politics of utopianism, of queerness as a horizon. Much of 21st century queer politics has been characterised, thus far, as one of pragmatic institutional integration — of straightening. In her criticism of Andrew Sullivan's emblematic neoliberal gay text *Virtually Normal*, Duggan offers a withering insight into the state of '*The New World Order! Coming soon to a mainstream near you*':

> ''equality' becomes narrow, formal access to a few conservatizing institutions, 'freedom' becomes impunity for bigotry and vast

> inequalities in commercial life and civil society, the 'right to privacy' becomes domestic confinement, and democratic politics itself becomes something to be escaped.' (Duggan 2002, 191)

There has been an '*erosion of the gay and lesbian political imagination*' if we find our mainstream politics fleeing from the radically re-configured futurism queerness offers: why do we talk about means of existence that offer no possibilities beyond those pre-determined by the dominating voices of the already-in-power as advances in our rights? (Muñoz 2009, 21) Indeed, a queerness that is '*not here yet*' is the queerness that drives my practice, a desire to '*strive, in the face of the here and now's totalising rendering of reality, to think and feel a then and there*' (Muñoz 2009, 1 — Muñoz's Emphasis)— a then and there in which queer bodies and narratives, regardless of how well they fit the dominant institutional narratives of the here and now, are taken into account. There is a need to develop escape routes from uncritical reflection and reification of the institutional control of bodies, in queer politics and in queer art practice, and that is the primary goal of this project. This will be explored in reactive composition: both through an investigation of how it can reproduce politics of control, and the development of queer methodologies to construct reactive systems with radical queer politics at their centre.

2.2. Power, Interaction, and Composition.

There is a necessity to begin the discussion of this project, therefore, by establishing how the power relations of interaction design and western compositional practice can be critically recontextualised in the context of queer understandings of power (Foucault 1972, 66). Firstly, it is necessary to interrogate a common site where the power relationships of reactive systems and music surface. This is explored in the techniques of mass-participation and feedback employed in marketing, both as a phenomenon that has shaped the design of reactive software systems, as discussed by Galloway and Thacker (2007), and how it has shaped the production and recording of music, as discussed by Attali (1985). Secondly, we will explore how the institutional sites responsible for these traceable effects delimit who is allowed to speak and what can be spoken. We will also address how this influences the construction of conceptual approaches to interaction and performance, and how the themes and theories of the discourses that emerge from this produce strategies of control. Finally, we will discuss how this project aims to develop new methodologies through expanding cracks in points of incompatibility within discourses of western compositional practice and interaction.

It is important to note that what is being analysed here are certain regularities of the discourses around interaction and composition (Foucault 1972, 72). This study does not aim to find or investigate the origins and genesis of these fields, but rather to try and account for and interrogate the networks of force relations that form these fields in an attempt to inform a queer approach to these fields.

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There are two primary models we can understand as structuring interactions between a system/its design and users. The first is what both Galloway and Thacker (2007, 122) and Paine (2002, 296) refer to as *interaction*. Interactive technologies form entire systems of distributed peers, in various groupings, in which each peer is able to physically affect the other. In interaction there is the necessity for reciprocal action and mutual experience, Paine suggests artificial intelligence as a means of achieving this in the context of human-computer interaction (Paine 2002, p 296). Developing earlier, but also continuing to develop in parallel with interaction is a process Galloway and Thacker refer to as *feedback* and what Paine describes as *reactive* work. For the purposes of this paper, we will understand *feedback* as a process undertaken by systems and their designers, and *reactivity* as a property of systems consisting of a monodirectional informatic loop that enables feedback, from producer/designer to consumer/user and back to producer/designer via the market, research, polling, data gathering, sensing of bodies, and so on. This informatic loop supplies a flow of information from the user to the monitoring system — designer, institutional, technological or otherwise — that drives decision-making based on the criteria selected to asses that information (Galloway and Thacker 2007, 122). As Galloway and Thacker (2007, 123) comment, in reactive systems 'the audience is structurally unable to achieve a symmetrical relationship of communication with the apparatus (no matter how loudly one yells back at the screen)'. Feedback is observed in the context of music through the mechanisms of creating, selling and recording. As Attali notes, profit under 20th century capitalism is found in the creation of the most perfect mould for repetition. In this sense, the creation of notation and recordings for repeated performance and playback can be understood as primarily being structured through a feedback process through the mechanisms of the market (Attali 1985, 118 and 128).

There are several key observations that can be made from these surface realities, however the most important for this project is that reactive systems rely on technologies of surveillance and data collection. These technologies inevitably generalise populations while also paying close attention to the detail of individual bodies through the lens of that generalisation. The process of feedback not only refines the product, but the institutional view of those consuming; certainly from a queer perspective there is an element of fear in being recognized as an outlier from this dataset by potentially oppressive technologies of the state. A methodology of continually emerging importance, biometrics, is a method of power-knowledge that privileges feature

extraction for the expansion of a data-set over a negotiated, self-identified engagement with a system. Indeed, the use of biology as a state method for identification and control of social sub-categories (class, sexuality, gender, etc.) as the basis for an understanding of human relations has been part of western discourse from Malthus (1798) and Darwin (1859) to Foucault (2010). This is echoed in the way recording has shaped the expectations of replicability in performance and composition. The performer's body is something to be trained to meet the standards of a pre-set ideal, the composer's work to expected to either enforce or increase these standards. Individual input is limited to a narrow field of interpretative choices within the scope of tradition and the composer's stated preferences.

Queer experience understands the necessity to intervene in the reification of these strategies of control and identification of the non-standard across discursive boundaries as a matter of life and death. There is an urgency to interrogating these practices, but also reconstruct existing practices with radically altered priorities and power relations. To do so, we need to examine from where control emanates in the existing structure. Galloway and Thacker argue that power in networks lies somewhere between the hierarchical powers enforced by the nation-state and those at the level of protocol: the laws of encoding, decoding and transmission that govern the functioning of the non-geographically bound (but by no means apolitical) networks. It is a result of a wide array of power relations and institutional prerogatives that reactive systems and scores present a narrow range of responses as the power to shape and negotiate your relationship to a system in a power relationship that is fundamentally asymmetric. This dramatically influences the construction of conceptual approaches to making software and to making art. The most obviously traceable surface effect in contemporary western compositional practice is the tensions raised in graphic scoring practice between the acts of composition and performance. Hope (2011) elegantly explores this in the context of programmers (who she earlier compares to performers [p.56]); acknowledging the compositional potential in those who realise conceptual work unfixes composition from the limits of a linear score. In interaction design, it is best epitomised in the aforementioned terminological tension between reactive and interactive. "The tension in both of these cases are uncomfortable truths of ingrained power relations. The notion of the "genius narrative" thrives on the presence of the all-knowing composer: there is no room for leaving open, or, worse still, *needing* input. One cannot describe one's system as *reactive* because it acknowledges the uncomfortable realities of participation in institutional powers under capitalism. Inevitably, these structures are replicated and formulate into strategies of control that enforce these underlying power relations.

The intervention into this that queer theory offers can shift these distinctions from binaries into something far hazier. Questions of a 'line' at which composition becomes performance or improvisation become irrelevant when decoupled from the power dynamics that demanded that distinction, and certainly reactive work that is decoupled from the power dynamics implied by a certain mode of designer/user relations can question what it means to interact with a system. Along with access to the code, and the intrinsic input my graphic notation suggests, this project explores the critical leverage of protocol to achieve this goal, channelling and interpreting input in a way that interrogates the social construction of interaction. Protocol acts on the *edges*, the connections that exist between *nodes* in a network. Following this line of thinking, composition takes on its more material definition, addressing the make-up of a reactive performance on the level of substrate: the 'wiring' — physical, software and cultural — between nodes. If notation is usually a surface effect of career spanning process of feedback, in this project it is both a critical stand-in for this tradition and, in the ways the reactive systems in this project disrupt and make this tradition unstable, a surface effect of the re-negotiated power relations operating on the level of protocol. This forms a critical engagement with the hierarchical power that enforces a model of the score as an ineluctable document of compositional genius, and with an assumption of the ability to record and repeat as a fundamental priority of composition. If an observable response from a system is usually a surface effect of a series of choices made exclusively by a designer designed to respond to measured and constant surveillance of a user, in this project it blurs perceptions of whether it was either designed, performed, or even a response in the first place. This forms a critical engagement with the hierarchical power that enforces a model of the designer as the only decision maker, and with an assumption that surveillance is fundamental to reactive systems.

The interactive and graphic scores that act as a technological backbone for the work in this project are a site of convergence between the theoretical background of this research and the strategies mobilised in my work to address the themes and theories unearthed in that process. They mobilise a critical illegibility that doubly reflects on the illegibility of queer bodies in cis-straight frames of reference and on the opacity of the technology on which we rely (Menkman 2011, 11). The vastness of potentiality of the network structure and the queer body, its illegibility and free-floating definition, make it impossible to exist in any other state of understanding.

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This project exists at the points of incompatibility in these discourses, between reactive and interactive, and between composition and performance. We have seen how queer theory offers critical insight into the power structures of interaction and composition, and this project mobilises this to further widen and interrogate

these cracks in discourses. The result is methodologies of co-creation, participation and sensing bodies that aim to reconstruct low-level and far reaching power relations in interaction. This context leaves us with rich terms for a practice-lead investigation into a queer approach to reactive composition, and for the development of software that enables different approaches to performance and composition that reflects these values.

3. THE SOFTWARE

Inrush is a continuation of a trend in my practice more broadly to explore openness and negotiability by performers/users. There was a conscious choice to use well documented, industry standard software in the hope that performers would either be more familiar with or easily able to familiarise themselves with the basic coding skills required to interact with the structural elements of the pieces. I decided to work with Max and Processing, which both also have their own Integrated Development Environments (IDEs), because they fulfil these requirements and are also both very stable and well-established languages.

A20 and *Scène* were chosen as case studies for this paper because they represent two approaches to score generation — algorithmic and through performance — but like all of my practice have a critical drive that I very much identify with my understanding of queerness. Even in *scène*, where the queerness is less opaque, as a composer it is important to me that my desire for negotiation and autonomy is something that comes directly from a queer experience of the body and the body's place in the world. While this is not an exclusively queer idea, for me it is the framing that is most intuitive and allows me to engage more fully in forming a criticism of how heteronormativity limits bodies in my work and in life.

3.1. General Notes

The score drawing software developed for what I am now calling *Inrush* was designed to draw and control the drawing of traditional and graphic notation using a combination of Open Sound Control (OSC) messages and pre-programmed phrases and controls. Written in Processing, a flexible and user-friendly programming language designed to teach coding skills to those in the visual arts, the code is highly legible and extensible by end users (Reas and Fry 2006). The software accepts OSC messages formatted similarly to standard MIDI for note duration, pitch and velocity, and custom messages for changing note scroll speed, the number of and OSC addressing of staves, player colours, and for drawing lines and curves.

The Max/MSP patches developed for the three pieces in this research are likewise extensible and flexible to suit end user needs. I've focussed on designing patches that are modular, but still show clear signal flow and design logic so they can be pulled apart and re-constructed by end-users as necessary, in the hope that this transparency will be encouraged more broadly in designing patches for performance.

3.2. A20

A20 is a generative piece for piano, projection, generative score and electronics. It's an exploration of outer-suburbia, the centre of Australia, queerness and constant motion. An audience member once jokingly remarked (thinking I was out of earshot) that the projection in this piece is a bit "Mad Max"; amusingly, it's precisely the inability to be identified and vastness of inner-Australia that drove me to make this piece, and to document my (semi-regular) drives between Adelaide and Sydney. There is something about the potentiality of being on the road in between two far-apart places that unsettles our definitional systems – if you're not in Adelaide or Sydney where are you? If you're not gay or straight what are you? The places in between exist, of course, but in a blurry infinity of choices; of course, between Adelaide and Sydney there are an infinity of blurred, queer spaces too, with illegible and obscured identities and desires.

Maybe the problem is the geography – pre-mapped positionalities of identity, desire and place. How can you perform your way out of the erasure implicit in this way of mapping? Massumi suggests that we get caught in cultural gridlocks like this because our ways of understanding bodies subtract movement from the equation. Even bodies that change position on the cultural grid are defined by their start and end points, the notion of a body in transition is lost to an understanding that privileges the convenience of pre-defined socio-cultural destinations (Massumi 2002, 3–5). *A20* uses this cultural stasis as an interactive framework, deliberately employing the erasure of blanket processing and exploring this to a point of hypertrophy through constant feedback in (potentially endurance) performance. The footage that is only identifiable as *passing through*, not able to be pinned down more than vaguely within a physical or definitional geography beyond this.

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The primary outcomes of this piece, in terms of moving towards establishing a queer reactive compositional practice, are related to exploring the power relationships taken for granted in the performance of notated music. The act of being watched and measured by technology is one that has broad social context. Blas writes about this in the context of 'Fag Face Mask', his work that addresses a Tufts University study towards biometric means of detecting gay faces, arguing darkness and illegibility are a queer strategy for survival in a political environment that uses technology to identify the un(re)productive in populations. (Blas 2012, 129) Blas uses the facial recognition data of many queer men,

mashed together and glitched to the point of becoming alien and unrecognizable by technology. This blatantly critiques the goals of the Tufts study, but also normative approaches to technology more broadly. Certainly, the mask itself is a twisted product of generalisation. As well as signifying a resistance to technology, it also represents the failure of efforts to distil humanity through generalising technological methodologies. But, of course, Blas' work is critical outside of its technological context; a deformed (almost like it was beaten), lurid mask the colour of a bruise has an immediate and prescient signification in the context of queer protest and police violence. I agree with Blas that there is a necessity to engage with techno-ethical issues in a material and nuanced way that reflects both on the technical and social implications of queer art making. In the case of *A20*, the point of interrogation is how technologies of ordering and control affect bodies, especially in the context of reactive work.

In A20, the software draws traditional notation, with weighted shading of the note-heads indicating dynamics, using midi data sent from Max/MSP in real-time. The max patch controls how the material is traversed through pre-built sub-patches and a modular, simple to re-construct interface. It sends OSC messages to Resolume Arena to change footage, analyses the audio and visual content of the current footage using Max's built-in FFT tools and Syphon for Jitter, converts this into pitch, velocity and audio processing information using the rules specified in the sub-patch in use, and finally outputs audio and OSC messages to the score-drawing software. This signal flow was designed to be as simple to intervene at every point by end-users, allowing performers with intermediate technical knowledge to shape how the piece is structured and interprets data.

The advantage of using my *Inrush* patches in this context was in the ability to quickly design performances that suited spaces. As my approach to the material involves heavy amounts of looping and, therefore, gradually a large amount of feedback, being able to quickly adapt the piece to fit different durations, spaces and instruments has been very helpful. Likewise, being able to make multiple algorithmic interpretations of the same material for different ensembles (for example, *Ensemble Offspring* in 2015) and have them still feel like the same piece has been a telling process in terms of understanding how material is prioritised in this piece and my practice (essentially: if there's feedback and footage, it feels like *A20*, and that leaves a broad scope for interpretation). As feedback is, in itself, a phenomenon that requires the relinquishing of control, so this patch and the freedom of the performer to cut up footage so intrinsically linked to my body a relinquishing of compositional control and an exercise in queer vulnerability.

As a starting point, we should investigate how and why performers interact with this material. Lingel and Naaman (2012) focussed on the exchange and sharing of self-filmed footage of live performances by bands, and how the sharing and distributing of lived experiences of musical performances reflected economies of gift giving and interpersonal play (p. 3). The shift from audiences 'waving their lighters… [to] mobile phones raised to capture video for rapid uploading to a variety of social media sites' has broader significance to other aspects of networked existence. (Lingel and Naaman 2012, 1) This articulation of interaction, a desire to shape and frame an individualised experience of material, has parallels with how composer-performer power relationships are renegotiated in this piece through the methodology with which the materials of the piece are structured. The system comes preloaded with hours of footage, prebuilt-means of processing it, and certainly plenty of room for implementing solutions to navigating it beyond those provided, yet – with no further structure – is outside of the general expectation of either a complete score, or an instruction on how to create a complete score. In this case, therefore, the performers are also themselves system designers, devising means of sonification of the materials, and flipping the power dynamics usually inherent in performing interactive work. This renegotiation of power dynamics extends further: the use of technology the performers participate in follows along similar lines of engagement in that, like the performers on Lingel and Naaman's stage, *my body* is what is primarily on the line. In her essay *Field Recording as a Performative Act*, Anderson notes that in field recording she 'perform[s her] body, and in these conversations caught in-the-moment, [her] voice'. (Anderson 2015) Indeed, in A20 I am present in the footage, both as the person operating the camera 90% of the time, but also because I am performing the action of filming. This is also true of the audio: my voice, me or my brother's choices in music, my brother's voice and engine noise are the core musical material. It is a unique vulnerability to be edited, manipulated and erased – but one that is key to the conceptual success of the piece.

This is the key point of renegotiation and critical interrogation of the relationship between performer/user and composer/designer: that my body is the primary subject of the control exerted in this piece. The result is an experience of my body, in this piece, that Massumi calls a *'body without an image'*: an accumulation of movements, intensities and meanings separated from the terms of their original existence. There is something about the material, the shifting, panning, zooming, bouncing camera is felt as something tactile and gestural, completely removed from the original form of its existence (my body) and felt as sensation. The exhaustion (and its anticipation) of a potentially endurance performance enforces this. To have this constant state of motion captured, freeze-framed and cut-up by technology is to create a tension between my body outside of linear time and the body of the pianist, a sort of phantom possibility that bleeds across from the screen to the performer and the audience – that this

fragmentation and erasure is an inevitability. (Massumi 2002, 57-63) This has two consequences. The first brings us back to Blas: we can resist (mis)representation and control of the queer body using counter-technological techniques, while still mobilising technology as a site of resistance against normative methodologies of control and a means of users having a sensation of queer ways of knowing. The second is ethical, and reveals the consequences of the power dynamics of reactive design. Given that the usual state of affairs in physical human computer interaction places the body of the user of the system at the mercy of the designer, this approach to reactive design – with two levels of design, and the body of the designer at the mercy of the user – has offered insight into the erasure, manipulation and decontexualisation of the body in reactivity. To address this directly is to address something felt keenly in a queer context. Erasure is a reality of queer living (Eisner 2013, 37), but also offers potential for resistance. To be drowned out by constant feedback and over-exposure, too, resonates with queer experience in a palpable and material way. Mimicry of institutional technologies of control is a sign of double articulation, a reflection of the limits imposed by strategies of control but also an appropriation of how this is visualised as a criticism of that power, making it hyperbolic and pushing it to a point where its inherent structural biases are exposed. (Muñoz 1999, 77–78) This mimicry has potential for both the communication of real emotion and real change in my body un-manipulable and un-maskable by technology, but also of emotion and change brought on precisely through the articulation of power-knowledge by reactive systems. (Massumi 2002, 64) On the protocological level, this is further affirmed by Menkman. The techno-cultural 'flow' that produces this sort of reactive design relies on the build-up of norms over long periods of time. The deviations from the standard flow in this piece capture 'the machine revealing itself', but also an 'infinity of possible alternative discourses, of other possible modes of explanation and interpretation', an unforeseen incomprehension in the face of the expected. (Kluitenberg 2002 in Menkman 2011, 29–30) This is an ethical void as much as it is an informatic void, the infinity is also one of countless articulations of power and control, and imaginings of how that plays out on the surface of a body without an image, felt physically by the audience and performers.

Like Blas' work, however, this work is not purely one felt as a product of technology. The experience of erasure, invisibility and control by distant forces in interaction is one that is echoed in the realities of outer suburban and rural Australia, and especially queerness in those contexts. As much as this piece is about an unmasking and renegotiation of technological power structures taken for granted, it is also about that drive from Elizabeth to Adelaide, and from New South Wales to South Australia. The technology encompasses the discomfort and conflict of this emotional landscape – tensions between representation and darkening, between identification and disidentification, between fragmentary sensation and the perceived wholeness of a lived experience – but also questions the necessity and ethics of imposing an interpretative frame on any of it. If any conclusion is to be drawn from A20, as a piece and as an experience, it is that the richness of 'open' work is in its infinity of pathways – the horror vacui that reveals the vast possibilities of experience. (Menkman 2011, 30)

3.3. scène

"the dominant, continuing search for a noiseless channel has been, and will always be no more than a regrettable, ill-fated dogma" (Menkman 2011, 1) *Scène* exists in a space between the disintegration of classical sensibilities, and their furthest abstraction; a broken digitisation of our music culture. *Scène* revels in this decay, moments of clarity breaking down and drifting apart; in and out of sync. A process of negotiation and listening; performance as utopia at the edge of the void. When I first scored *scène* on paper, I was interested specifically in how traditional notation could be broken with glitch techniques, and how the notation of electronics could intrude on spaces reserved for acoustic instruments. In the version presented at this conference, I was interested in how I could explore new ways of renegotiating composition through a digital platform in rehearsal, allowing quick feedback and experimentation in rehearsal and performances that *fit* performers.

In *scène*, the flutist improvises from a paper score, and the electronic musician both performs audio-processing of the flutist and the generation of the score, distributing blocks of improvisatory material to player in the ensemble ad-libitum in rehearsal, with the possibility to move towards more strictly timed interpretations by performance so they can focus on audio-processing. The max-patch comes pre-loaded with the modular score materials, and some tools for manipulating them, and the scoring classes used can draw simple graphics and blocks of traditional notation. Much of this came out of the experience of working as a copyist on Cat Hope's 2017 Opera *Speechless* and, therefore, with a combination of Decibel Score Player and Adobe Illustrator The end product was a fairly polished, negotiated and collaborative effort with all the hallmarks of the composer's best work, but it was also a gruelling and time-expensive process. *Scène's* simple musical materials, which rely on their combination and dispersal for complexity, were perfect testing grounds for renegotiating a workflow that met both the desire to explore and combine the creative input of musicians and the composer in a way that was timely, intuitive, and fully embraced a queer futurity I aim to instil in my software and my works that explore improvisation.

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As of the time of writing, the key outcome of adapting *scène* with *Inrush* has been the ability to critically re-address expectations of clarity in electronics. Indeed, the use of loopers, white noise, and sine waves in the electronics part was a deliberate choice in terms of aural recognition, and pushing them and the sound of the flute to points of critical hypertrophy key to pushing against the limitations of the idea of the technological "neutrality" of the electronics and the instrument. In the context of reactive technology, as I've previously outlined, the emotional and political stakes are different and the criticism more important. Using a method of scoring that has the potential to be as "clear" as processing power allows one to explore notions of clarity and ambiguity outside of our power to process. In the context of semi-improvised music, and especially semi-improvised music engaged with queerness, this feels absolutely primary.

4. CONCLUSION AND FUTURE DIRECTIONS

Inrush aimed, first and foremost, to develop theoretical and technological approaches for making queer interactive work The research and programming associated with this project has laid an important foundational methodology for queer interactive work in my practice through a critical exploration of reactive paradigms. The technological and theoretical insights this research provides on a small-scale hint at rich territory for future research into how queerness can be used to shape critical technologies. The implications of this research will be far reaching in my future research and artistic practice, and I look forward to consolidating my code and patches into something that broadens the potential for collaboration and co-creation across art-forms and disciplines.

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